

Abdelrahman Ihab

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🌐 LinkedIn | 🐙 GitHub | 🌐 Portfolio

Professional Summary

Fourth-year Computer Engineering student at E-JUST specializing in AI, Computer Vision, Cybersecurity, and Robotics. Led computer vision for an award-winning ROV that placed 1st at MATE ROV 2026 Pioneer Level (+ Best Design & Best Presentation). Designed and deployed end-to-end AI systems for malware detection, real-time sports analytics, and autonomous edge surveillance. Holds a 120-hour Fortinet Cybersecurity certification (93%), IBM SkillsBuild Cybersecurity, and Google Cybersecurity credentials. Proven track record across 5 industry internships in network administration, cybersecurity, and IT operations. Seeking internship or research roles in AI engineering, computer vision, or offensive/defensive cybersecurity.

Education

Egypt-Japan University of Science and Technology (E-JUST)

Oct. 2022 – Present

Bachelor of Science in Computer Engineering

- **AI & Intelligent Systems:** Machine Learning, Deep Learning, Computer Vision
- **Security & Systems:** Embedded Systems, Operating Systems, Parallel & Distributed Computing
- **Core Engineering:** Algorithms, Data Structures, Theory of Computation, Computer Networks, Robotics

Alexandria International School

Sep. 2008 – Jun. 2022

IGCSE, Grade: 99.5%

Experience

Vortex Robotics

Jan. 2026 – Present

Computer Vision / ML Engineer

- Designed and deployed a full YOLOv8 computer vision pipeline for an underwater ROV competition, achieving over 95% trajectory continuity.
- Built a European Green Crab detection system using synthetic data augmentation to overcome limited real-world data, enabling reliable biodiversity analysis in underwater environments.
- Developed a 3D coral garden reconstruction model with real-world spatial calibration and accurate metric scaling for scientific reporting.
- Team secured **1st Place**, the **Best Design Award**, and the **Best Presentation Award** at the MATE ROV 2026 Pioneer Regional Competition, qualifying for the international competition in Canada.

National Telecom Institute (NTI) – ITIDA

Aug. 2025 – Sep. 2025

Fortinet Cybersecurity Trainee

- Completed a 120-hour intensive training program (30 hours soft skills, 90 hours technical) with an overall score of **93%**.
- Configured FortiGate firewalls including security policies, NAT, FSSO authentication, routing protocols, and FortiManager integration.
- Implemented SSL VPN, IPSec VPN, SD-WAN, Security Fabric, IPS, Antivirus, Web Filtering, and High Availability solutions.
- Studied cryptography, PKI, certificate lifecycle management, and intrusion prevention in production-simulated environments.

Information Technology Institute (ITI)

Sep. 2025 – Oct. 2025

Artificial Intelligence Trainee – NVIDIA AI Program

- Completed **Building LLM Applications with Prompt Engineering**, covering prompt engineering, LLM optimization, and API integration for Large Language Model applications.
- Completed **AI for All: From Basics to Generative AI**, covering AI fundamentals, model training, deep learning concepts, and Generative AI applications.

Ezz Dekheila Steel Company (EZDK)

Jul. 2025 – Aug. 2025

IT Internship Trainee

- Gained enterprise IT operations experience within one of Egypt's largest industrial organizations.
- Applied networking, cybersecurity, and web development skills to real production infrastructure and IT workflows.
- Observed and assisted with corporate network management, cloud computing practices, and project coordination.

PETROJET

Aug. 2025

Computer Engineer Intern

- Diagnosed and resolved network device failures across enterprise infrastructure, maintaining system uptime and reliability.
- Gained hands-on experience with SAP enterprise systems and collaborated with cross-functional IT teams.
- Documented network configurations and troubleshooting procedures to support operational knowledge transfer.

GIECO – Gharably Integrated Engineering Company

Jul. 2024 – Sep. 2024

Network Administrator Intern

- Assisted in planning, configuring, installing, and maintaining enterprise network infrastructure.
- Configured and maintained routers, switches, and VPN endpoints while monitoring network performance.
- Supported WAN and LAN deployment across multiple office locations.

ETHYDCO

Jul. 2024

Cybersecurity Intern

- Identified and documented security vulnerabilities across industrial control systems and proposed mitigation measures.
- Conducted vulnerability assessments and contributed to incident monitoring, threat detection, and response activities.
- Assisted in implementing authentication controls and access management policies aligned with cybersecurity best practices.

Projects

Nautix | 1st Place, Best Design, Best Presentation – MATE ROV 2026

- Engineered a YOLOv8 object detection pipeline for European Green Crab identification using synthetic data augmentation and custom training to maximize accuracy on limited real-world data.
- Implemented trajectory interpolation for occluded frame recovery, achieving over **95%** trajectory continuity across underwater test dives.
- Developed a 3D coral garden reconstruction system with calibrated real-world metric scaling for scientific biodiversity reporting.
- **Tech Stack:** Python, YOLOv8, OpenCV, NumPy, Synthetic Data Generation, Homography, 3D Reconstruction
GitHub: github.com/Abdelrahman-288/Mate-ROV-Pioneer-2026

VLM-Enhanced Autonomous AI Security Patrol | Graduation Project

- Designed and developed an autonomous AI-powered security patrol system combining TinyML and Vision-Language Models (VLMs) for intelligent edge surveillance.
- Implemented a hierarchical AI pipeline on Arduino Portenta H7 using FOMO object detection with the Groq Vision-Language API for efficient edge-cloud inference.
- Optimized an INT8-quantized TinyML model achieving **86.9%** detection accuracy while significantly reducing memory usage and cloud inference costs.

- **Tech Stack:** Arduino Portenta H7, Portenta Vision Shield, TinyML, FOMO, Groq Vision-Language API
- **GitHub:** github.com/Abdelrahman-288/VLM-Enhanced-Autonomous-AI-Security-Patrol

Hybrid AI Malware Classification & Detection System | *Deep Learning & Cybersecurity*

- Architected a four-stage malware classification pipeline using binary visualization, CNN feature extraction, opcode analysis with BiLSTM, and ensemble fusion on the Microsoft BIG 2015 dataset (10,868 malware samples across 9 families).
- Performed static malware analysis using PE header inspection, opcode extraction, entropy analysis, and behavioral feature engineering with pefile and Ghidra.
- Integrated Grad-CAM explainability and deployed an interactive Streamlit dashboard for real-time malware prediction, visualization, and automated threat reporting.
- **Tech Stack:** Python, TensorFlow, CNN, BiLSTM, Ghidra, pefile, Streamlit, Grad-CAM, NumPy, Pandas
- **GitHub:** github.com/Abdelrahman-288/malware-classification-system

Tennis Match Analytics | *Real-Time Sports Analytics*

- Fine-tuned YOLOv8 for player and ball detection on fixed-camera video, improving ball detection Average Precision (AP) by **18%** over baseline.
- Applied homography-based court mapping and trajectory interpolation, achieving **93.31%** accuracy on computed match statistics.
- Developed a custom CNN for court keypoint regression, eliminating manual calibration while generating player movement, shot direction, and rally analytics.
- **Tech Stack:** Python, YOLOv8, PyTorch, OpenCV, CNN, Homography, Keypoint Regression
- **GitHub:** github.com/Abdelrahman-288/Tennis-Analysis

Deep Face Detection System | *Transfer Learning*

- Built a custom face detector by fine-tuning a pretrained VGG16 backbone with bounding-box regression and binary classification heads for face localization.
- Created and annotated a custom dataset using OpenCV and LabelMe with VOC and COCO export pipelines.
- Deployed the trained model as an interactive Gradio web application supporting real-time image and webcam inference.
- **Tech Stack:** Python, TensorFlow, VGG16, OpenCV, LabelMe, Gradio
- **GitHub:** github.com/Abdelrahman-288/Deep_Face_Detection_Project

Heart Disease Prediction | *Machine Learning*

- Developed a complete machine learning pipeline including preprocessing, exploratory data analysis, PCA, and feature selection using the UCI Heart Disease dataset.
- Trained and optimized Logistic Regression, Decision Trees, Random Forest, and Support Vector Machine models using GridSearchCV and RandomizedSearchCV.
- Applied unsupervised clustering to identify patient subgroups and deployed the solution through an interactive Streamlit application.
- **Tech Stack:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Streamlit
- **GitHub:** github.com/Abdelrahman-288/Heart_Disease_Prediction

Certifications & Credentials

• MATE ROV 2026 – Pioneer Level Regional Champion — MATE ROV Competition	Apr. 2026
• Universal Robots UR20/30 eLearning — Universal Robots Academy	Apr. 2026
• Hands-on Introduction to Linux Commands & Shell Scripting — IBM, Coursera	Feb. 2026
• Embedded Linux Course — IEEE	Dec. 2025
• AI & Machine Learning Foundations — Sprints	Aug. 2025
• IBM SkillsBuild Cybersecurity Certificate — IBM SkillsBuild	Aug. 2025
• English Language Certificate — Oxford	Aug. 2022